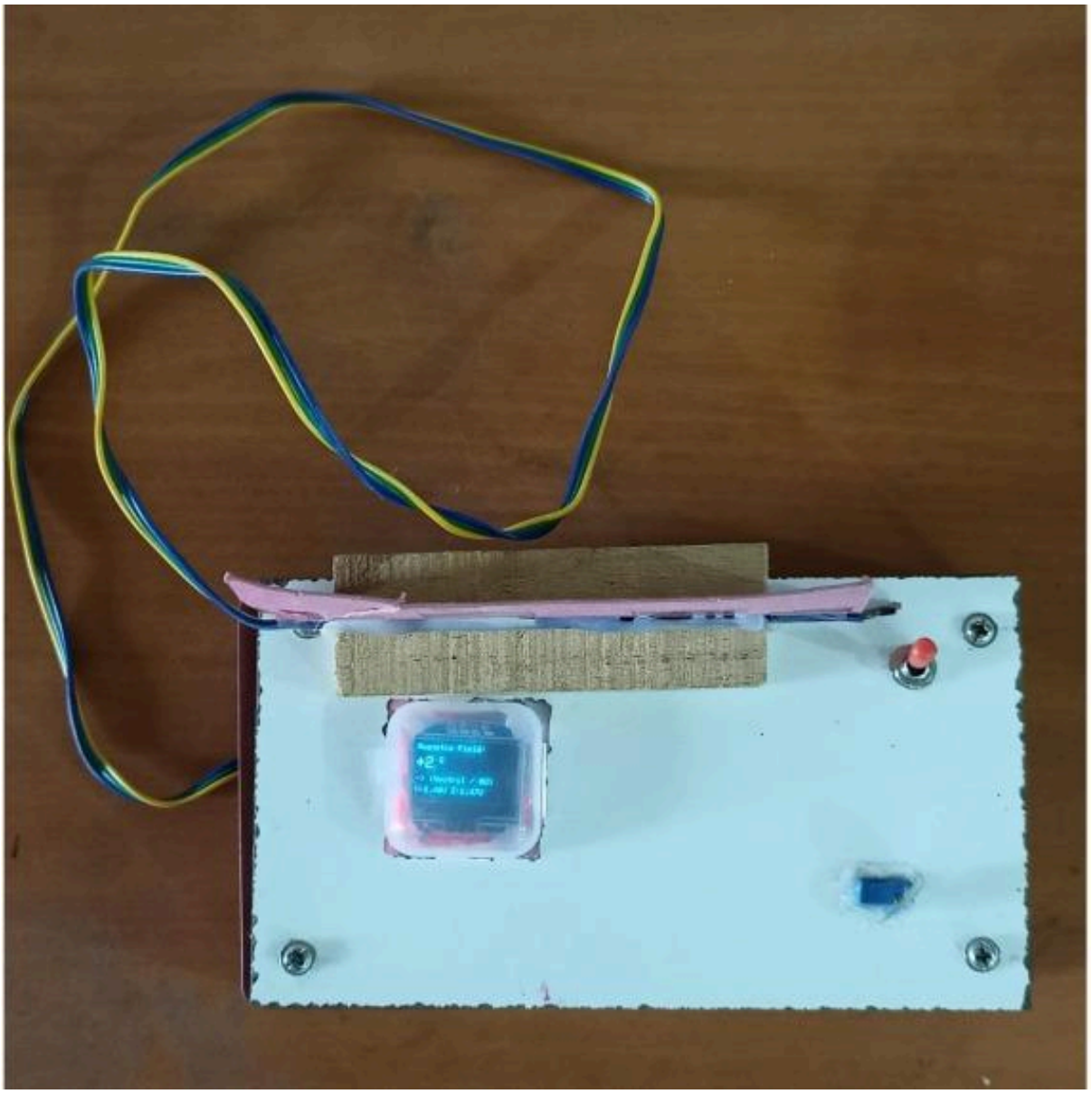
GAUSS METER

Magnetic Field Measurement Using Hall Effect Sensor

Objective :-

To design and develop a low-cost digital Gauss meter using an Arduino Nano and an analog Hall-effect sensor to measure magnetic field strength in real time. The measured magnetic field strength and polarity are displayed on a 0.96-inch I²C OLED display. The system is designed to provide a simple, portable, and user-friendly method for educational demonstrations, laboratory experiments, and basic magnetic field measurement and mapping.



Required Components

• **Arduino Nano** – Main microcontroller used to process the Hall-effect sensor readings and control the OLED display.
• **Analog Hall-Effect Sensor (A1302 / SS49E / similar)** – Used to detect and measure the magnetic field strength.
• **0.96-inch OLED Display with I²C Module** – Used to display the measured magnetic field strength and polarity.
• **Potentiometer (10 kΩ)** – Used for calibration/adjustment of the sensor readings.
• **Wooden Strip** – Used as a linear guide or rail for moving the Hall-effect sensor along a fixed path during magnetic field measurement.
• **Jumper Wires** – Used to make electrical connections between the components.
• **Breadboard or PCB (Optional)** – Used for assembling and testing the circuit.
• **USB Cable** – Used to power and program the Arduino Nano.
• **Magnets** – Used as test sources for generating magnetic fields.

Hardware Connections

Component	Pin	Arduino Nano Pin	Arduino Nano Pin
Hall Sensor (SS49E)	VCC (Pin 1)	5V	Looking at the labeled face
	GND (Pin 2)	GND	Center pin
	OUT (Pin 3)	A0	Linear analog signal
Potentiometer	Pin 1	5V	
	Pin 2 (Wiper)	A1	Zero-point reference (V_zero)
	Pin 3	GND	
I2C Display (OLED/LCD)	VCC	5V	
	GND	GND	
	SDA	A4	Hardware I2C Data
	SCL	A5	Hardware I2C Clock

Working Principle:

Hall Effect Sensor:

• Senses the perpendicular magnetic flux density (B).
• Provides a proportional analog voltage output corresponding to the magnetic field strength.
• Outputs a quiescent baseline reference voltage (V0 approx 2.5 V) in the absence of a magnetic field (0 G).
• Voltage increase above V_0 indicates a South Pole, while voltage drop below V_0 indicates a North Pole.

Arduino Processing:

• Reads raw 10-bit analog values (0 – 1023) from the sensor (A0) and calibration potentiometer (A1).
• Converts the analog reading to actual voltage:

$$\text{Voltage} = \frac{\text{analogRead} \times 5.0}{1023.0}$$

• Calculates the magnetic flux density in Gauss using:

$$\text{Gauss} = \frac{\text{Voltage} - V_0}{\text{Sensitivity}}$$

Evaluates the sign and magnitude of the Gauss value to classify magnetic polarity (North, South, or Neutral).
• Renders real-time Gauss values, polarity status, and baseline voltages on the 0.96" I2C OLED Display, while simultaneously streaming telemetry to the Serial Monitor.

Extensions and Improvements:

• You can add a live waveform graph on the OLED or computer screen to see field strength changes in real time.
• You can connect a Bluetooth module like HC-05 to send and save Gauss data directly to a smartphone.
• You can add a single push-button to automatically zero and calibrate the sensor without turning a potentiometer.
• You can use an external 16-bit ADC module to make the readings much smoother and detect
tiny magnetic variations.

• You can upgrade to a 3-axis Hall sensor to track magnetic fields in 3D space rather than just
one direction.
• You can add a temperature sensor to fix measurement drift when electromagnets or sensors
get warm during testing.

Applications:

• It helps students easily see magnetic fields in action during physics and science classroom demonstrations.
• You can instantly check whether a magnet is North or South and measure its exact surface strength.
• It lets you map and track how magnetic strength changes at different distances along a sliding rail.
• It serves as a handy, budget-friendly tool for testing sensors in DIY electronics and robotics projects.
• It provides a reliable setup for conducting fundamental magnetic field experiments in student laboratories.

Conclusion

The developed Gauss Meter measures magnetic field strength in real time using a Hall-effect
sensor and Arduino Nano. The 0.96-inch I²C OLED display shows the magnetic field value and
polarity, while the potentiometer provides zero-point calibration. Overall, the project is low-cost,
compact, and useful for educational and experimental applications.
